ICRA 2026 BARN Challenge

Join us on Slack



Wondering how the state-of-the-art navigation systems work on autonomous navigation in highly-constrained spaces? Check out our detailed report on the results and findings of The BARN Challenge at ICRA 2026 here!

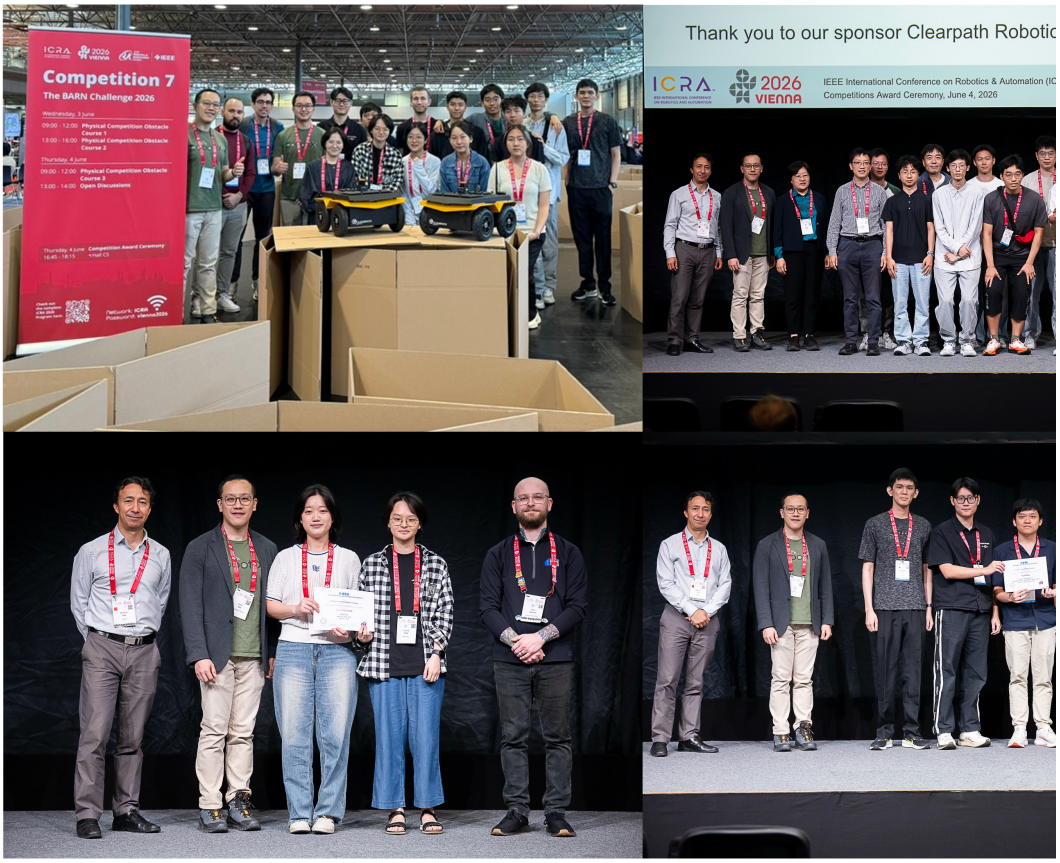
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Congrats IN^{2}BOT, EW-Glab, and Team Robo for winning the 1st, 2nd, and 3rd place of The BARN Challenge at ICRA 2026! See you next year!



About

Designing autonomous robot navigation systems has been a topic of interest to the robotics community for decades. Indeed, many existing navigation systems allow robots to move from one point to another in a collision-free manner, which may create the impression that navigation is a solved problem. However, autonomous mobile robots still struggle in many scenarios, e.g., colliding with or getting stuck in novel and

tightly constrained spaces. These troublesome scenarios have many real-world implications, such as poor navigation performance in adversarial search and rescue environments, in naturally cluttered daily households, and in congested social spaces such as classrooms, offices, and cafeterias. Meeting these real-world challenges requires systems that can both successfully and efficiently navigate the environment with confidence, posing fundamental challenges to current autonomous systems, from perception to control. Therefore, the Benchmark Autonomous Robot Navigation (BARN) Challenge aims at creating a benchmark for state-of-the-art navigation systems and pushing the boundaries of their performance in these challenging and highly constrained environments.

At ICRA 2025, nine teams participated in the simulation qualifier of The Forth BARN Challenge in Atlanta, with eight being invited to and four attending the physical finals. We congratulate RRSL, RobotiXX, and UVA AMR for winning the 1st, 2nd, and 3rd place of The BARN Challenge at ICRA 2025. We published the results and findings about how the state-of-the-art navigation systems work in highly-constrained spaces in IEEE Robotics and Automation Magazine.

At ICRA 2024, The Third BARN Challenge in Yokohama had six teams competed in the simulation competition and top four teams in the physical finals. We congratulate LiCS-KI,

MLDA_EEE, and AIMS for winning the 1st, 2nd, and 3rd place of The BARN Challenge at ICRA 2024. We published the results and findings about how the state-of-the-art navigation systems work in highly-constrained spaces in IEEE Robotics and Automation Magazine.

At ICRA 2023, we successfully continued with our second BARN Challenge in London. Ten teams competed in the simulation competition and the top six teams advanced into the physical finals. We congratulate KUL+FM, INVENTEC, and University of Almeria for winning the 1st, 2nd, and 3rd place of The BARN Challenge at ICRA 2023. We published the results and findings about how the state-of-the-art navigation systems work in highly-constrained spaces in IEEE Robotics and Automation Magazine.

At ICRA 2022, we successfully launched the first BARN Challenge in Philadelphia. Five teams competed in the simulation competition and the top three teams advanced into the physical finals. We congratulate UT AMRL, UVA AMR, and Temple TRAIL for winning the 1st, 2nd, and 3rd place of The BARN Challenge at ICRA 2022. We published the results and findings about how the state-of-the-art navigation systems work in highly-constrained spaces in IEEE Robotics and Automation Magazine.

The Challenge

The BARN Challenge will take place primarily on the simulated BARN dataset and also physically at the conference venue in Vienna.

The BARN dataset comprises 300 pre-generated navigation environments, ranging from easy open spaces to difficult highly constrained ones, and an environment generator that can generate novel BARN environments. The task is to navigate a Clearpath Jackal robot from a predefined start to a goal location as quickly as possible without any collision. The Jackal robot will be standardized with a 2D LiDAR, a motor controller with a max speed of 2m/s, and appropriate computational resources. Participants will need to develop navigation systems which consume the standardized LiDAR input, run all computation onboard using the provided resources, and output motion commands to drive the motors. Participants are welcome to use any approaches to tackle the navigation problem, such as using classical sampling-based or optimization-based planners, end-to-end learning, or hybrid approaches. The following infrastructure will be provided by the competition organizers:

- The 300 pre-generated BARN environments
- The BARN environment generator to generate novel environments
- Baseline navigation systems including

classical (default DWA, fast DWA, and E-Band), end-to-end learning (e2e RL and LfLH), and hybrid (APPLR) approaches
- A training pipeline running the standardized Jackal robot in Gazebo simulation with Robot Operating System (ROS) Melodic (in Ubuntu 18.04), with the option of being containerized in Docker or Singularity containers for fast and standardized setup and evaluation
- Standardized evaluation pipelines for ROS and ROS2 to compete against other navigation systems

Competition Rules

During the competition, another 50 new BARN evaluation environments will be generated, which will not be accessible to the public. Each participating team is required to submit the developed navigation system as a (collection of) launchable ROS node(s). The final performance will be evaluated based on a standardized metric that considers navigation success rate (collision or not reaching the goal count as failure), actual traversal time, and environment difficulty (measured by optimal traversal time). Specially, the score s for navigating each environment i will be computed as

$$s_i = 1^{\text{success}} \times \frac{\text{OT}}{\text{clip}(\text{AT}_i, 2\text{OT}_i, 8\text{OT}_i)}$$

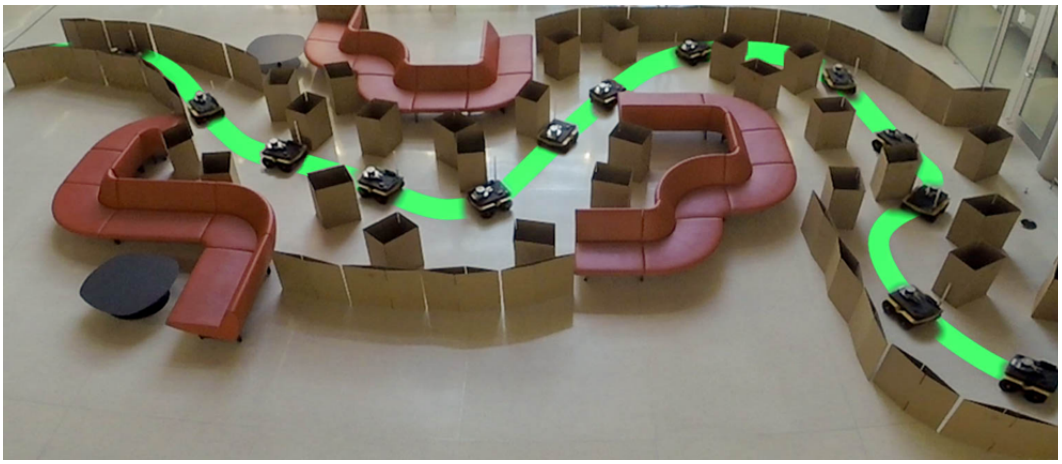
where the indicator 1^{success} is set to 1 if the robot reaches the navigation goal without any collisions, and set to 0 otherwise. AT denotes the actual traversal time, while OT denotes the optimal traversal time, as an indicator of the environment difficulty and measured by the shortest traversal time assuming the robot always travels at its maximum speed ($2\text{m}/\text{s}$):

$$\text{OT}_i = \frac{\text{Path Length}_i}{\text{Maximum Speed}}$$

The Path Length is provided by the BARN dataset based on Dijkstra's search. The clip function clips AT_i within 2OT_i and 8OT_i, in order to assure navigating extremely quickly or slowly in easy or difficult environments won't disproportionally scale the score. **Notice that we changed the upper performance bound 4OT_i from the past two years to 2OT_i this year, considering the improving navigation performance.** The overall score of each team is the score averaged over all 50 test BARN environments, with 10 trials in each environment. Higher scores indicate better navigation performance.

During the in-person conference in London, physical obstacle courses will be set up at the conference venue using cardboard

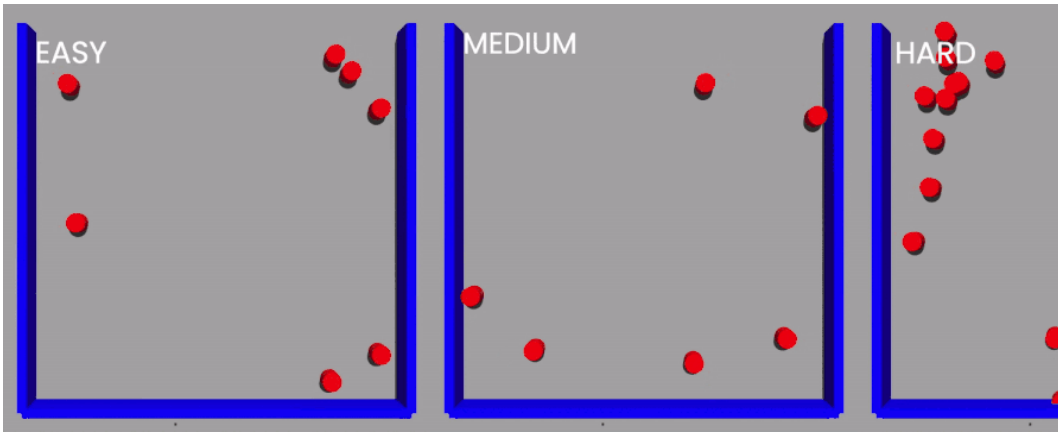
boxes (see example below). A physical Clearpath Jackal robot will be provided by the competition sponsor, Clearpath Robotics, with the same standard sensor and actuator suites as in the simulation. However, a sim-to-real gap may still exist due to factors such as friction coefficient between physical wheels and venue floor, electric motor impedance and resistance, real-world sensor noise distribution, etc. The top three teams will be invited to compete in the physical competition. In case a team is not able to attend the conference in person, the organizers will run their submitted navigation stack on their behalf on the physical robot at the conference. The team who achieves the highest collision-free navigation success rate and shortest traversal time wins the competition.

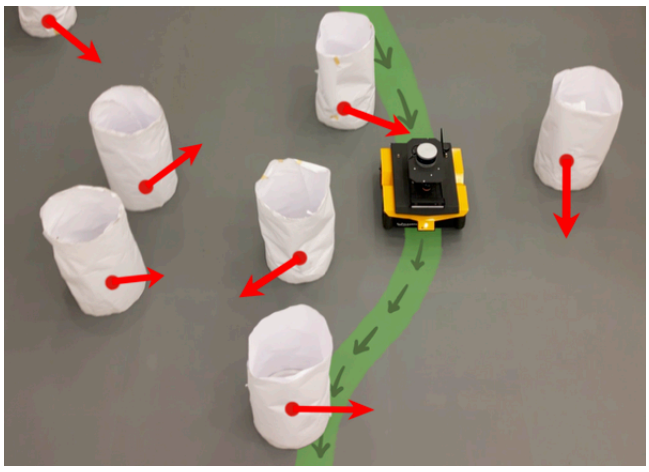


Due to the overhead of the simulator in the simulation phase, we do not specifically constrain computation. We will use a computer with Intel Xeon Gold 6342 CPU @ 2.80 GHz to evaluate all submissions. However, during the physical competition,

robot onboard computation will be limited to an Intel i3 CPU (i3-9100TE or similar) with 16GB of DDR4 RAM. GPUs are not available considering the small footprint of the LiDAR perception data.

Starting this year, we will introduce dynamic obstacles in our evaluation pipeline, both in the DynaBARN simulation environments and in physical test courses with iRobot Create robots acting as dynamic obstacles. As the first competition with dynamic obstacles, we will only reward good performance when facing dynamic obstacles in terms of extra bonuses, but not penalize poor performance. More details to come soon.





Leaderboard

Real World

2026

Ranking	Team	Success / Total Trials	Code Link
1	IN^{2}BOT	7/9	GitHub
2	EW-Glab	5/9	GitHub
3	Team Robo	3/9	GitHub

2025

Ranking	Team	Success / Total Trials	Code Link
1	RRSL	7/9	GitHub
2	RobotiXX	3/9	GitHub
3	UVA AMR	1/9	GitHub

2024

Ranking	Team	Success / Total Trials	Code Link
1	LiCS-KI	6/9	GitHub
2	MLDA_EEE	5/9 (78.8s)	GitHub
3	AIMS	5/9 (108.8s)	GitHub

2023

Ranking	Team	Success / Total Trials	Code Link
1	KUL+FM	9/9	GitHub
2	INVENTEC	6/9	GitHub
3	University of Almeria	5/9	GitHub

2022

Ranking	Team	Success / Total Trials	Code Link
1	UT AMRL	8/9	GitHub
2	UVA AMR	4/9	GitHub
3	Temple TRAIL	2/9	GitHub

Real-world challenge includes three different obstacle courses. Each team has 30 minutes to finish five timed trials. Top three trials are counted. The team that finishes the most successful trials (reaching the goal without any collision) wins. In the case of a tie, the team with the fasted average traversal time wins.

Simulation

The scores and ranking have been updated using the new scoring metric (previous $4\mathrm{OT}_i$ vs. current $2\mathrm{OT}_i$, moving the previous performance upper bound from 0.25 to the current 0.5). See The BARN Challenge 2023 Leaderboard for the previous scores using the old metric.

Ranking	Team	BARN (DynaBARN) Score	Comment
1	IN^{2}BOT	0.4975 (0.0000)	Beijing Institute of Technology
2	EW-Glab	0.4880 (0.0000)	Ewha Womans University
3	FSMT^{25}	0.4878 (0.0000)	NA
4	RobotiXX^{25}	0.4873 (0.1042)	George Mason University
5	KKato	0.4856 (0.0000)	Independent Researcher
6	UFAM + INDT	0.4814 (0.0000)	Universidade Federal do Amazonas + Institute of Technology
7	LiCS-KI^{24}	0.4762	Korea Advanced Institute of Science and Technology
8	EW-Glab^{25}	0.4736 (0.0000)	Ewha Womans University
9	AIMS^{24}	0.4723	The Hong Kong Polytechnic University
10	KUL+FM^{23}	0.4641	KU Leuven & Flanders Make
11	ASU ROAR	0.4619 (0.0000)	Ain-Shams University, Faculty of Engineering

12	LiCS-KI[25]	0.4611 (0.1125)	Korea Advanced Institute of Science and Technology
13	UT AMRL[22]	0.4594	The University of Texas at Austin
14	MDP-DWA[25]	0.4525 (0.0000)	Univerisity of Maryland
15	Team Robo	0.4515 (0.1236)	Nanyang Technological University, Singapore
16	MLDA-NTU[25]	0.4510 (0.0201)	Nanyang Technological University, Singapore
17	University of Almeria[23]	0.4460	University of Almeria & Teladoc
18	Helios	0.4455 (0.0000)	Independent Researcher
19	LfLH[b]	0.4354	Details
20	CRAL	0.4289 (0.0000)	University of Virginia
21	INVENTEC[23]	0.4206	Inventec Corporation
22	e2e[b]	0.4083	Details
23	Temple TRAIL[22]	0.4039	Temple University
24	EIT-NUS[24]	0.3795	Eastern Institute of Technology, Ningbo
25		0.3710	Staxel

	Staxel[23]		
26	PANAV	0.3682 (0.0000)	Indian Institute of Technology Patna
27	RRSL[25]	0.3558 (0.0001)	Michigan Technological University
28	UVA AMR[25]	0.3475 (0.0000)	University of Virginia
29	UVA AMR[23]	0.3475	University of Virginia
30	Deptron	0.3392 (0.0000)	Probabilistic Robotics Research Group, Yildiz Technical University
31	UVA AMR[22]	0.3256	University of Virginia
32	eYantra	0.3119 (0.0000)	ERTS Lab, Indian Institute of Technology Bombay
33	APPLR-DWA[b]	0.3052	Details
34	E-Band[b]	0.2832	Details
35	MLDA_EEE[24]	0.2476	Nanyang Technological University, Singapore
36	Yiyuiii[22]	0.2376	Nanjing University
	The MECO		

37	Barners[23]	0.2214	KU Leuven
38	Fast DWA[b]	0.2047	Details
39	NavBot[22]	0.1733	Indian Institute of Science, Bangalore
40	Default DWA[b]	0.1656	Details
41	Laboratory for Aero-Mechanical Autonomy (LAMA)	0.0517 (0.0000)	Indian Institute of Technology, Bombay
42	UT AMRL[23]	Missing	The University of Texas at Austin
43	Temple TRAIL[23]	Missing	Temple University
44	RIL[23]	Missing	Indian Institute of Science, Bengaluru
45	Armans Team[25]	NA	Pabna University of Science and Technology
46	Lumnonicity[23]	NA	AICoE, Jio
47	Tartu Team[24]	NA	University of Tartu
48	CCWSS[24]	NA	Wuhan University of Science and Technology
49	LoneBot	NA	Independent Researcher
			Multi Agent

50	VeRan	NA	Systems Lab, IISc Bangalore
51	TUBE	NA	University of Virginia
52	Team K	NA	SWM
53	PUST_BARNARS	NA	Information and Communicatoion Engineering

Red denotes 2026 participants.

[25] denotes 2025 participants.

[24] denotes 2024 participants.

[23] denotes 2023 participants.

[22] denotes 2022 participants.

[b] denotes baselines.

Participation

We have standardized the entire navigation pipeline using the BARN dataset in a standardized Singularity container. You should only modify the navigation system, and leave other parts intact. You can develop your navigation system in your local environment or in a container. You will need to upload your code (e.g., to GitHub) and submit a Singularityfile.def file which will access your code. We will build your Singularity container using the Singularityfile.def file you submitted, run the built container for evaluation, and then publish your score on our website. More detailed instructions can be found here. Please use this Submission Form to submit

your navigation stack to be evaluated by our standardized evaluation pipelines for ROS and ROS2.

Citing BARN

If you find BARN useful for your research, please cite the following paper:

```
@inproceedings{perille2020benchmarking,
title = {Benchmarking Metric
Ground Navigation},
author = {Perille, Daniel and
Truong, Abigail and Xiao, Xuesu
and Stone, Peter},
booktitle = {2020 IEEE
International Symposium on
Safety, Security and Rescue
Robotics (SSRR)},
year = {2020},
organization = {IEEE}
}
```

Schedule

Date	Event
January 1 2026	Online submission open
April 8 2026	Soft Submission Deadline[1]
April 22 2026	1st Round Physical Competition Invitations
May 8 2026	Hard Submission Deadline[2]
May 15 2026	2nd Round Physical

	Competition Invitations
June 1 2026 8a-6pm	Organizer Set-up
June 2 2026 8a-6pm	Team Set-up
June 3 2026 9a-noon	Physical Competition Obstacle Course 1
June 3 2026 1p-4p	Physical Competition Obstacle Course 2
June 4 2026 9a-noon	Physical Competition Obstacle Course 3
June 4 2026 1p-2p	Award Ceremony and Open Discussions

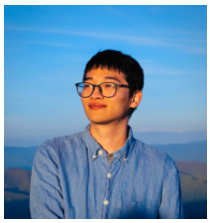
[1]Deadline to guarantee an invitation to the final physical competition given good navigation performance in simulation.

[2]Deadline to participate in the simulation competition. However, final physical competition eligibility will depend on the available capacity and travel arrangement made beforehand.

Organizers



Xuesu Xiao
GMU



Zifan Xu
UT Austin



Saad Abdul Ghani
GMU



Aniket Datar
GMU



Daeun Song
GMU



Peter Stone
UT Austin
Sony AI

Clearpath Robotics
Competition Sponsor

Links

- BARN Dataset
- BARN Paper
- BARN Video
- BARN Presentation at SSRR 2020
- Source Code to Customize BARN
- Source Code to Run Simulation Trials
- Example Simulation Trial Results
- The BARN Challenge 2025 Report
- The BARN Challenge 2024 Report
- The BARN Challenge 2023 Report
- The BARN Challenge 2022 Report

Contact

For questions, please contact:

Dr. Xuesu Xiao
Assistant Professor
Department of Computer Science
George Mason University
3401 Fairfax Dr, Arlington, VA 22201, United States
xiao@gmu.edu